

Source: Calculated with data from *Wall Street Journal* Market Data Center.

The other important detail in the exhibit is the column showing the implied portfolio bond allocation needed to build a STRIPS ladder of increasing length from a portfolio of \$1 million. The column reveals that every year of income from the ladder requires an allocation of about 4 percent of the portfolio, though this percentage declines at longer maturities where the yields are higher. With a \$1 million portfolio, the initial withdrawal rate to get \$40,000 of spending is 4 percent. The full retirement ladder could only support a 4 percent initial spending rate (with the 2 percent COLA for spending) for twenty-eight years, as the portfolio would be fully depleted in year twenty-nine.

The ladder cost is scalable. If you want \$100,000 of income, the cost is equal to the cost of \$40,000 of income times the ratio of the new income goal: $100,000 / 40,000 = 2.5$. Also, this exhibit works to show the cost for building a one-time bond ladder of any length up to thirty years. The total cost for a shorter ladder is simply the cost shown at that point in the cumulative cost column.

Simplified Assumptions for Bond Ladders

I have used a few simplifying assumptions to build these bond ladders. First, they are based on the wholesale asking prices for bonds on the secondary market published in the *Wall Street Journal*. These wholesale prices are for institutional trades of at least \$1 million. Household investors may have to pay an additional premium above these prices when purchasing individual bonds in a brokerage account.

Second, for each year, I used the bond with the earliest maturity date within that year rather than trying to optimize more strategically among multiple maturing issues.

Third, for TIPS there are years with in which no TIPS mature due to changes in the Treasury auction schedule. For those years, I buy additional shares of the closest previous maturity date available and assume that those funds grow at a 0 percent real rate until the year they